

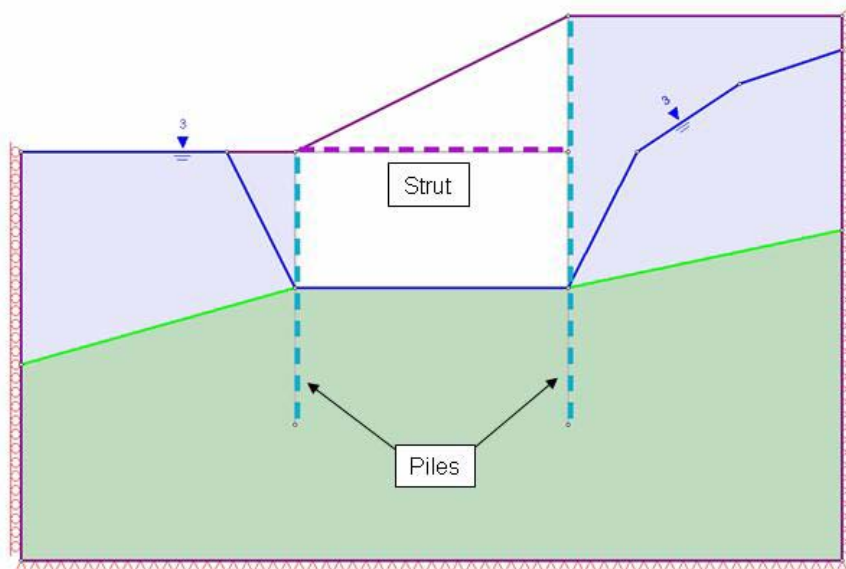
Trench Tutorial

In this tutorial, *Phase2* is used to simulate the excavation of a trench into a sloped embankment. The trench is supported by soldier piles and struts. The model is built in five stages:

1. Bring soil layers to equilibrium
2. Install a pile on the high side of the trench
3. Excavate the first section of the trench and lower the water table
4. Install a pile on the low side of the trench and install a strut between the two sides
5. Excavate the final section of the trench and lower the water table

Topics covered

- Soldier piles
- Strut with cross-sectional area and moment of inertia
- Staged support
- Staged piezometric lines
- Staged excavation
- Automatic Liner Removal option = OFF
- Show Values
- Plot Liner data



Model

We will start this tutorial by importing a file in which geometry, materials and boundary conditions have already been assigned. Start the *Phase2* Model program. Select File > Recent Folders > Tutorials Folder and open the file called **Tutorial 17 boundaries.fez**. You will see a model that looks like this:



This is Stage 1 of the model showing the soil layers prior to excavation.

Project Settings

The first thing to do is change the number of stages in the model. Go to the Analysis menu and choose Project Settings. Set the Number of Stages to 5. Click OK to close the dialog.

Automatic Liner Removal Option

Before we proceed with the modeling, go to the **Support** menu, and notice the **Automatic Liner Removal** option.

In order to model strut support in *Phase2*, the Automatic Liner Removal option must be turned OFF. This allows you to excavate the material on both sides of the strut (liner), without automatically removing the support. We will be doing this in Stage 5.

For the file you have just read in, the Automatic Liner removal option has already been turned off. (When the option is ON a checkmark appears beside the menu option, when it is OFF, no checkmark appears). Make sure the option is OFF. For more information about the Automatic Liner Removal option, it is recommended that you read the *Phase2* Help.

Stage 1 – Initial Conditions

For Stage 1 we only need to add a piezometric line to delineate the water table. Go to the Boundaries menu and choose Add Piezometric Line.

Enter the following coordinates:

0 , 6 (existing vertex)

4 , 6 (existing vertex)

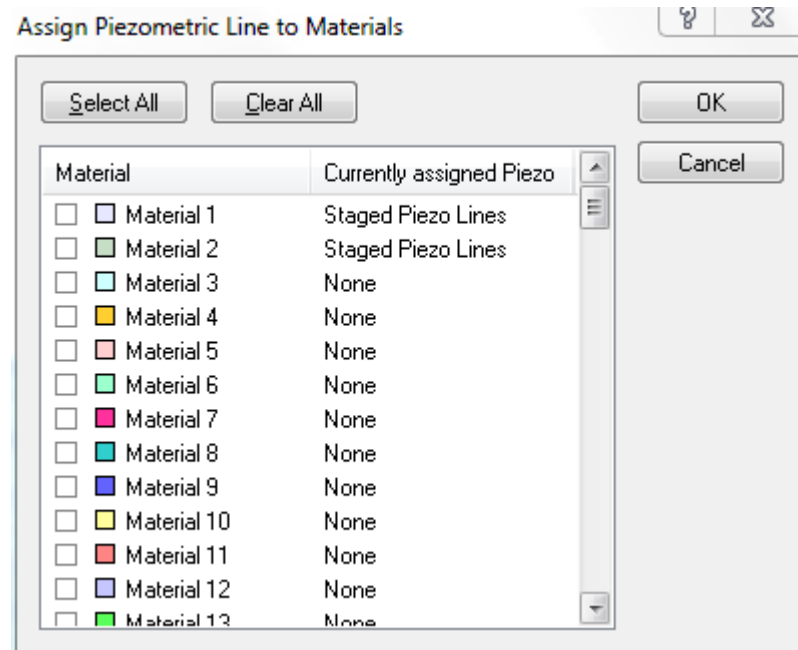
8 , 7.5

12 , 7.5

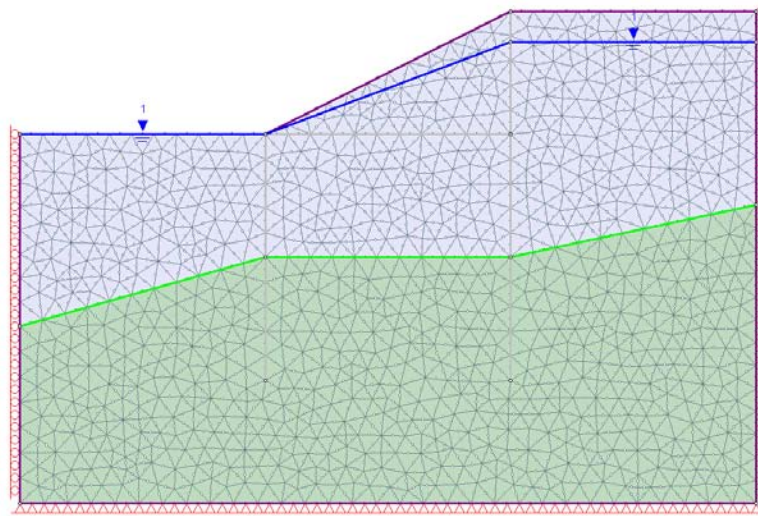
Hit Enter to finish entering the points.

NOTE: For existing vertices it is easier just to click on them than enter the numbers – if the cursor doesn't snap to existing vertices then right click and turn on the Snap options. For the final point you also will not have to type it in. The cursor should snap to the right boundary making a horizontal line.

You will now see a dialog asking to which materials you would like to assign the piezo. Click on the check boxes beside Material 1 and Material 2 as shown.



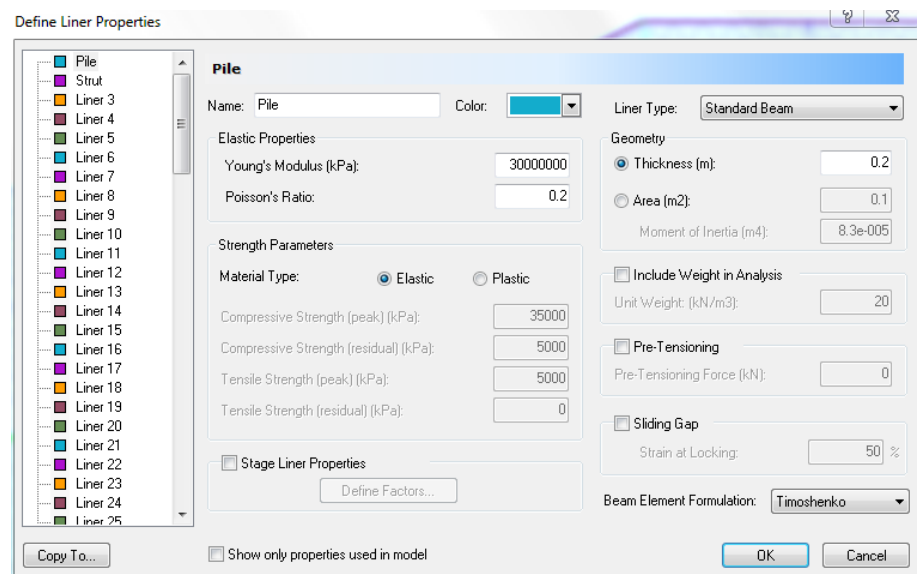
Click OK. Your model for Stage 1 should now look like this:



Stage 2 – Install pile support

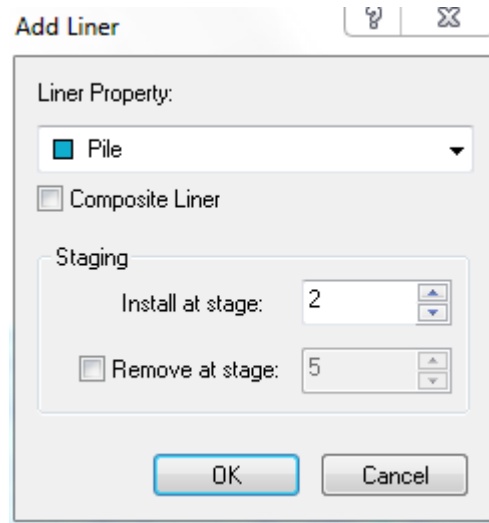
Select the Stage 2 tab. In Stage 2 we will add a pile to the right side of the future trench. We will implement this as a liner element. Another possible approach would be to implement this as a structural interface with a joint on either side of the liner allowing slip to occur between the liner and the soil. See Additional Exercises at the end of this tutorial.

First we need to set the properties of the pile. Select **Define Liners** from the Properties menu. Change the name of Liner 1 to Pile. Change the thickness to 0.2 m as shown.



Click OK to close the dialog.

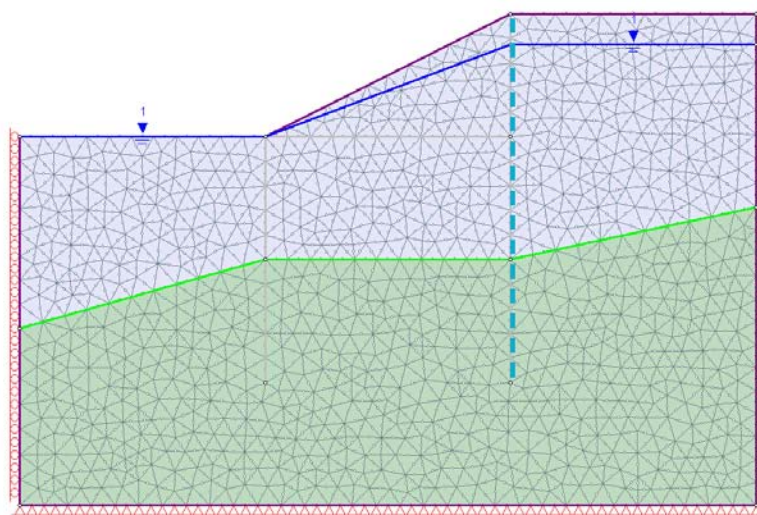
Install the liner by selecting Add Liner from the Support menu. You will see the Add Liner dialog as shown.



Ensure that the Liner Property is Pile and **Install at stage** is set to 2. Click OK.

Now select the three stage boundaries to the right of the future trench at $x = 8$ m. Hit Enter to finish selecting boundary segments.

Your model should look like this for Stage 2.



Stage 3 – Excavate soil and lower piezometric line

Click on the Stage 3 tab. In this stage we will excavate the first part of the trench and lower the water table.

Select Assign Properties from the Properties menu. Make sure that Materials is selected from the pull-down menu at the top. Now click on Excavate. Click in the triangular section at the top-middle of the model to excavate the first part of the trench. Close the Assign dialog.



We now need a new piezo line that runs below the new surface. Click the Add Piezo Line button. Enter the following points:

0 , 6 (existing vertex)

8 , 6 (existing vertex)

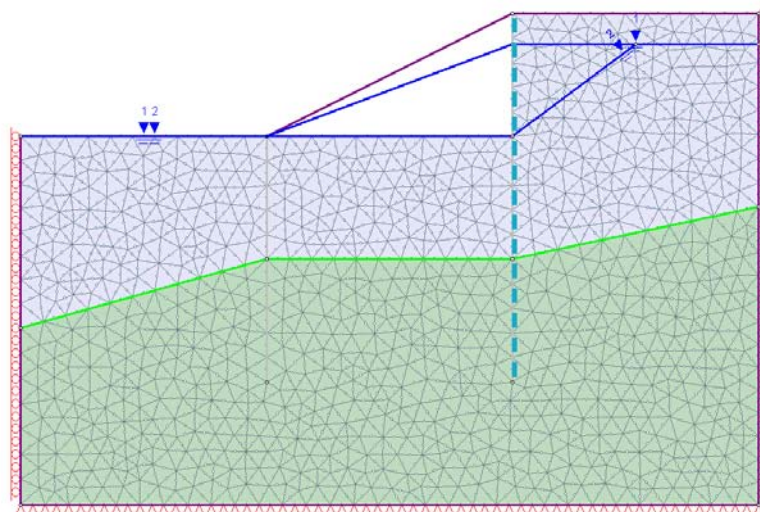
10 , 7.5

12 , 7.5 (existing vertex)

Hit Enter to finish entering points.

In the Assign Piezometric Line to Materials dialog, do not check any of the materials. The materials need to be assigned different piezo lines at different stages. We will sort this out in Stage 5. Click OK to close the dialog.

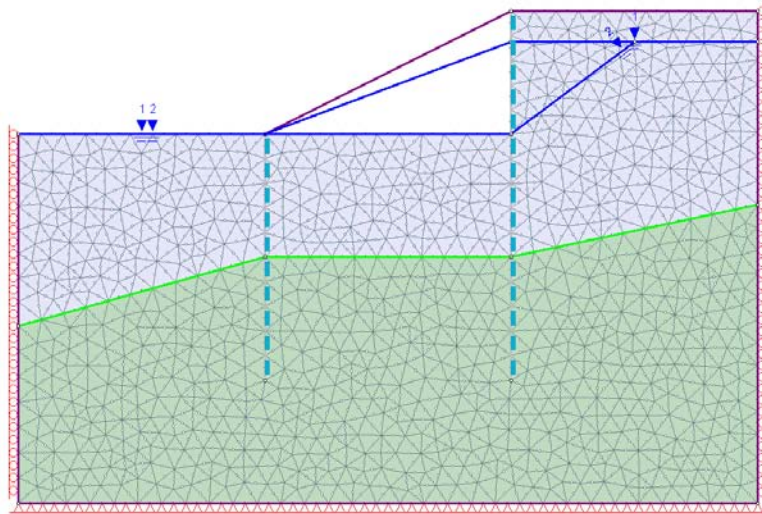
You will now see both piezo 1 and piezo 2 plotted. We need to stage the piezo lines but it will be easier to do this at the end of model building when all of the piezo lines are in place. For now just leave the two piezo lines as shown. The model for Stage 3 should look like this.



Stage 4 – Install second pile and strut support



Click on the Stage 4 tab. The properties of the pile have already been set up so we only need to install it. Click on the Add Liner button. Ensure the Liner Property is Pile and Install at Stage is set to 4. Click OK. Now select the stage boundaries to the left of the trench at $x = 4$ m as shown.



We will now install a strut between the two piles to support the excavation that will occur in Stage 5. First we need to set the properties of the strut. Click the Define Liner Properties button on the toolbar. Click on the tab for Liner 2. Change the name to Strut. Under Geometry, turn on the Area option. Change the Area to 0.2 m^2 and the Moment of Inertia to $1\text{e-}20 \text{ m}^4$ as shown.

Define Liner Properties

Strut

Name: Color:

Liner Type:

Elastic Properties

Young's Modulus (kPa):

Poisson's Ratio:

Strength Parameters

Material Type: ☒ Elastic ☐ Plastic

Compressive Strength (peak) (kPa):

Compressive Strength (residual) (kPa):

Tensile Strength (peak) (kPa):

Tensile Strength (residual) (kPa):

☐ Stage Liner Properties

Geometry

☐ Thickness (m):

☒ Area (m²):

Moment of Inertia (m⁴):

☐ Include Weight in Analysis

Unit Weight: (kN/m³):

☐ Pre-Tensioning

Pre-Tensioning Force (kN):

☐ Sliding Gap

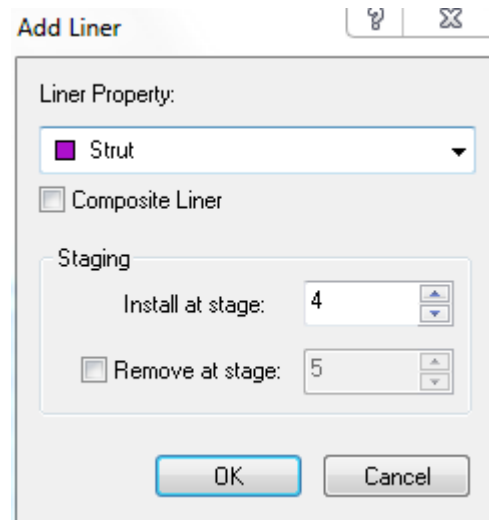
Strain at Locking: %

Beam Element Formulation:

☐ Show only properties used in model

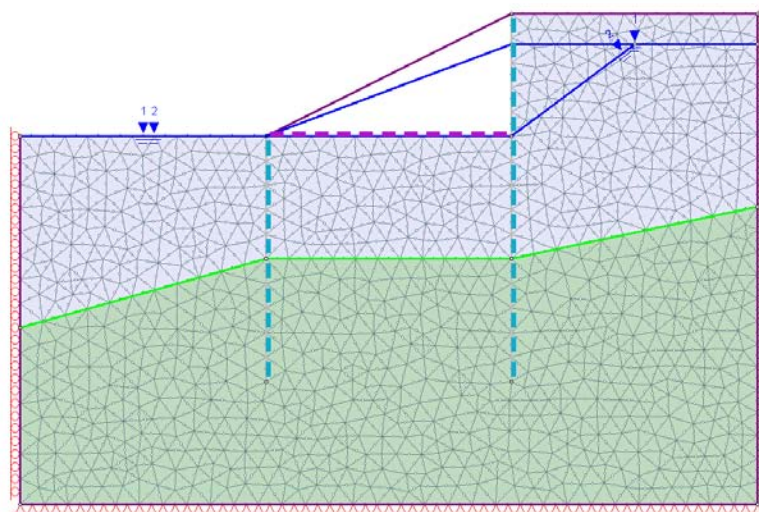
Setting a very low moment of inertia ensures that the beam will experience no flexure or torsion – only axial forces.

Now add the strut by selecting Add Liner from the Support menu. In the Add Liner dialog, select Strut from the pull-down menu for the Liner Property. Ensure that the Install at Stage option is set to 4.



Click OK.

Now click on the boundary segment at the top of the soil between the two piles. Right click and choose Done Selection. Your model should now look like this:



TIP: You may wish to hide the piezo lines prior to installing the strut. To do this right click anywhere in the *Phase2* window and choose Display Options. Now uncheck Piezometric Lines. You will also need to hide the piezo lines if you wish to access the liner strut information by right clicking on it, since the strut is “underneath” piezo line 2.

Stage 5 – Excavate soil and lower piezometric line

Click on the tab to show Stage 5. In this stage we will excavate the remainder of the trench and add another piezo line.



Click on the Assign Materials button. Click the Excavate button in the dialog and then click inside the lower section of the trench between the piles and just below the strut. Close the Assign dialog.

NOTE: because the **Automatic Liner Removal** option is turned OFF, this allows us to excavate the material on both sides of the strut, without automatically removing the strut. See the section earlier in this tutorial for more information.

We now need to add another piezo line below the new soil surface. From the Boundaries menu select Add Piezometric Line. Enter the following points:

0 , 6 (existing vertex)

3 , 6

4 , 4 (existing vertex)

8 , 4 (existing vertex)

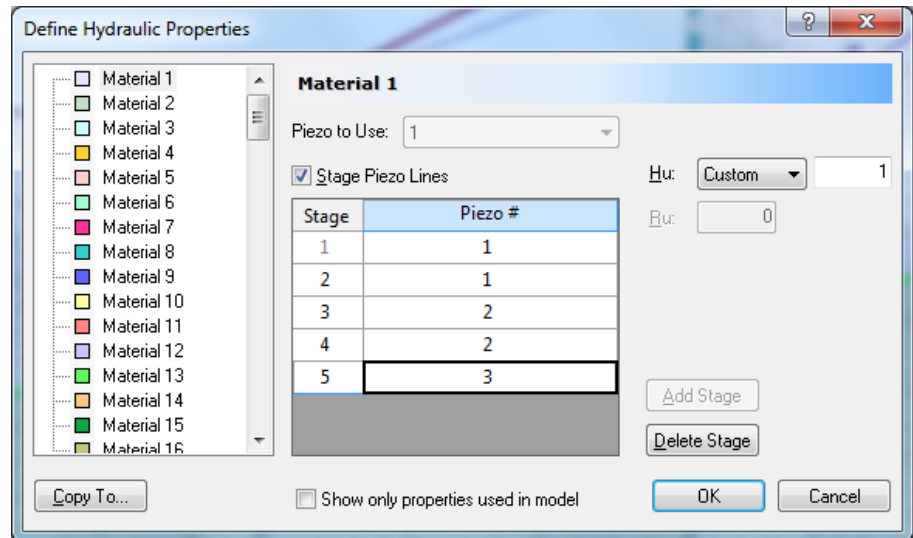
9 , 6

10.5 , 7

12 , 7.5 (existing vertex)

Hit Enter to finish entering points. In the resulting dialog, do not select any materials and click OK.

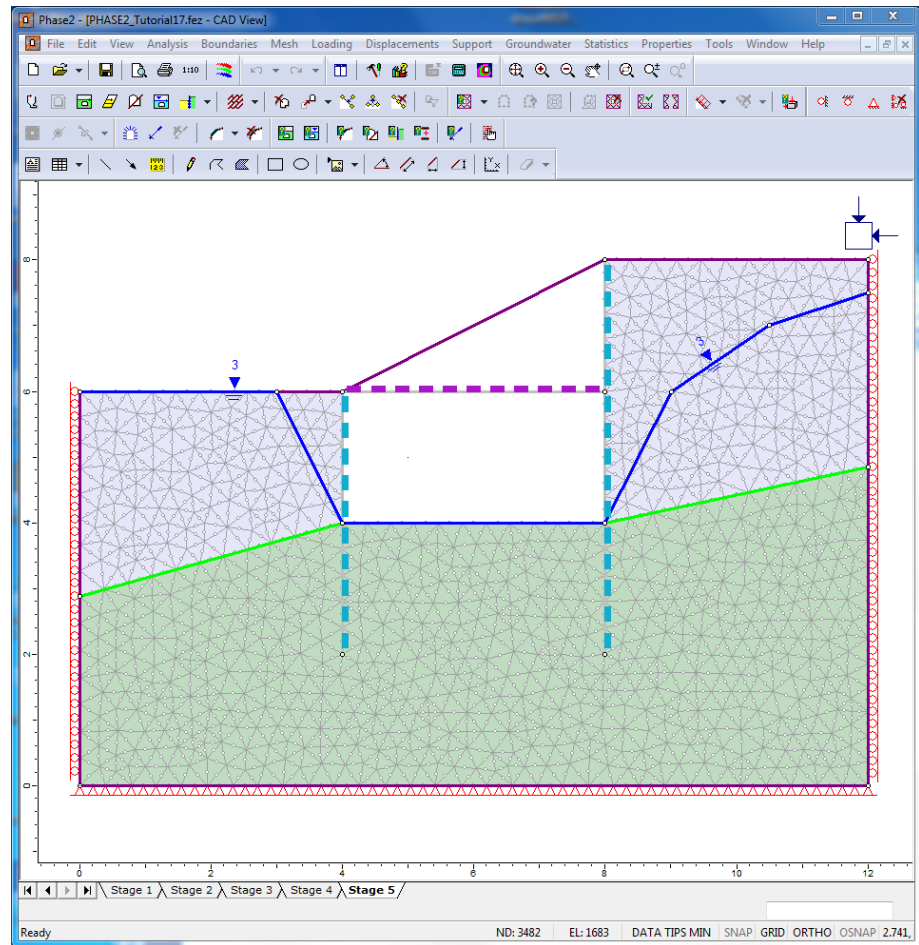
It is now time to assign the piezo lines to the correct stages. From the Properties menu select Define Hydraulic. For Material 1, turn on the option for Stage Piezo Lines. For Stage 1, set the Piezo # to 1. For Stage 2, set the Piezo # to 1. Now click the Add Stage button. For Stage 3 set the Piezo # to 2. Click Add Stage 2 more times to add stages 4 and 5. Set the Piezo # for Stage 5 to 3. The dialog should now look like this.



Material 2 is affected by the piezo lines in the same way as Material 1. Therefore click on the Material 2 tab and repeat the above steps to stage the piezo lines. TIP: you can also use the Copy To button in the dialog to copy the properties from one material to another. Click OK to close the dialog.

NOTE: You do not need to define the piezo # for every stage. If a stage is not listed it will automatically adopt the piezo # of the previous stage. Therefore in this example you could have only specified stages 1, 3 and 5. This can be done by setting up only 3 rows in the table and changing the stage numbers by clicking on the stage numbers in the first column.

You will still see 3 piezo lines on your model. To only see the piezo line for the relevant stage, go to the Groundwater menu and choose View Piezos by Stage. Your model should now look like this for Stage 5.



Click through the other stages to ensure that the piezo lines are where they should be. You have now completed the definition of the model. Save the model with a different name using the Save As option in the File menu.

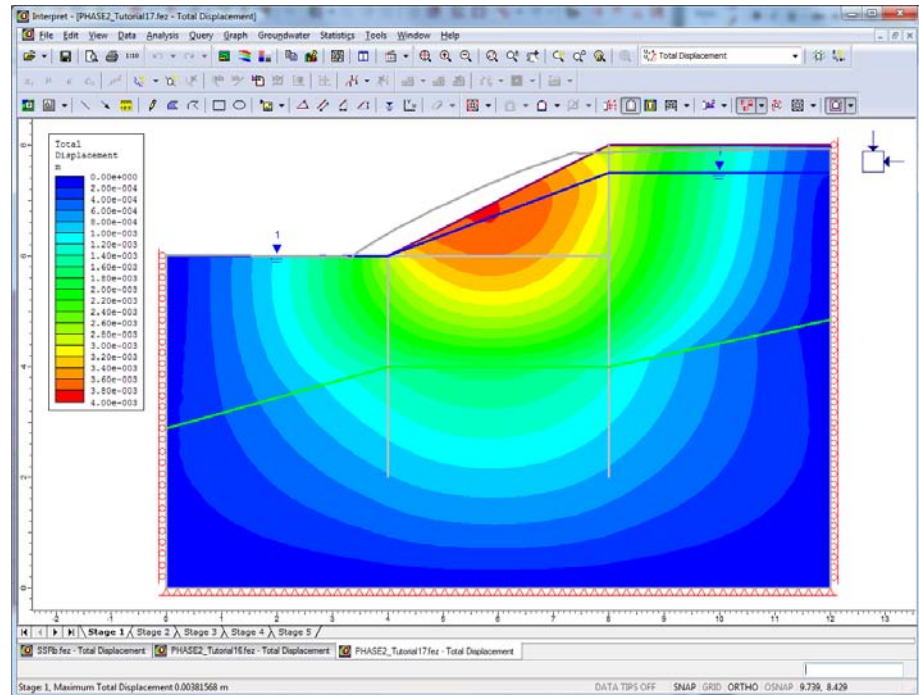
Compute

Run the model using the Compute option in the Analysis menu. The analysis should take a few minutes to run.

Once the model has finished computing (Compute dialog closes), select the Interpret option in the Analysis menu to view the results.

Interpret

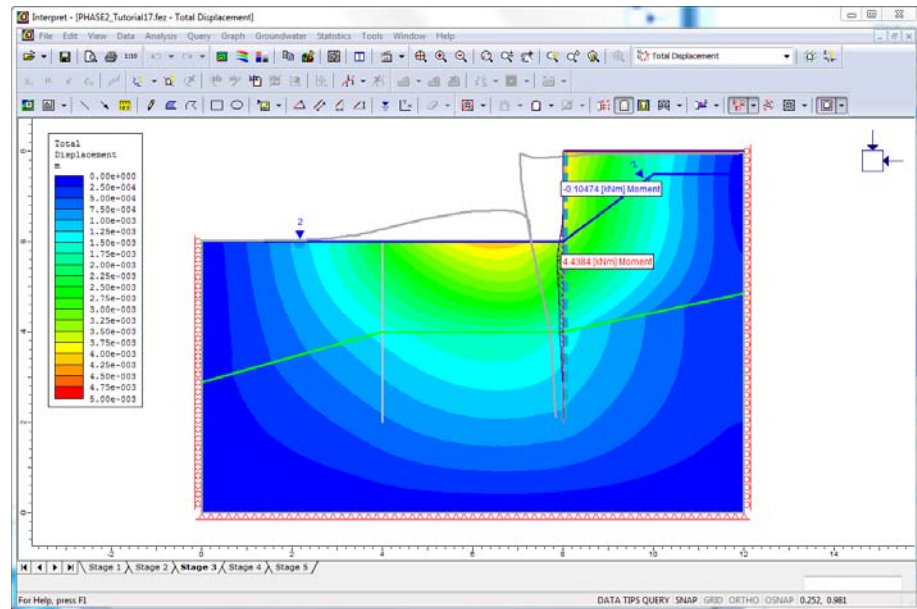
After you select the Interpret option, the Interpret program starts and reads the results of the analysis. You will see the maximum stresses for Stage 1. Change the contours to Total Displacement and turn on the deformed boundaries by clicking the Display Deformed Boundaries button. The plot should look like this:



You will see that some deformation has occurred already. If you press the Display Yielded Elements button, you will see that significant failure has occurred at the top of the slope.

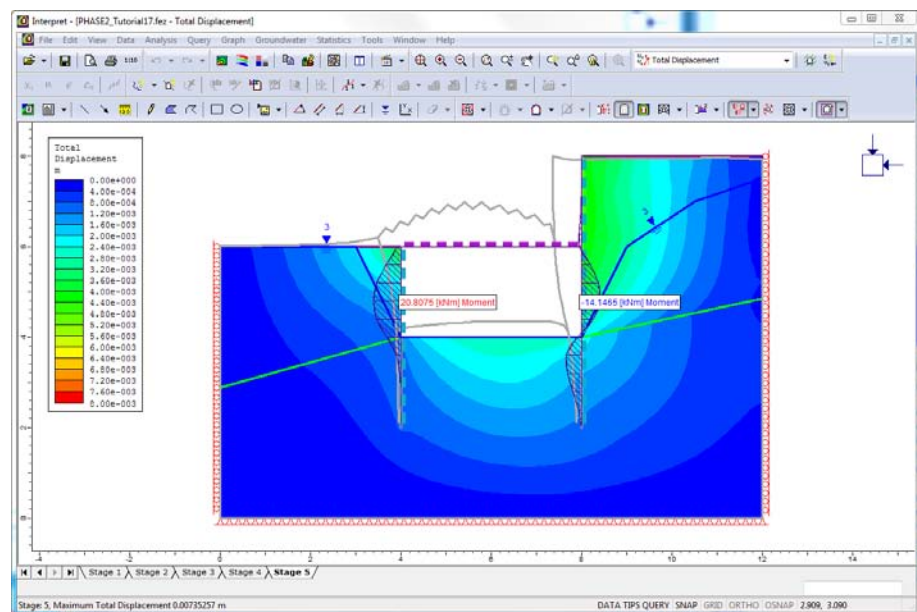
Turn off the yielded elements and click on the Stage 2 tab. You will see that the addition of the soldier pile has no effect until the trench is excavated.

Click on the Stage 3 tab. You will now see significant displacement at the right side and bottom of the trench. The pile appears to be bending due to the weight of the soil it is holding back. Verify this by right clicking on the liner and selecting Show Values > Bending Moment. Your plot should now look like this:



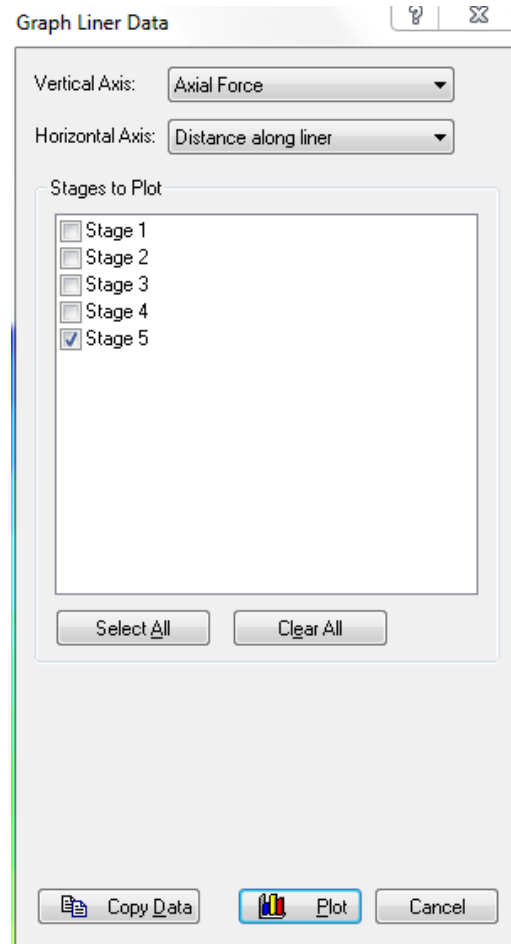
You can see positive bending moments just below the surface of the trench.

Click on the Stage 4 tab. You should see very little difference as the other pile and the strut are added. Click on the tab for Stage 5.

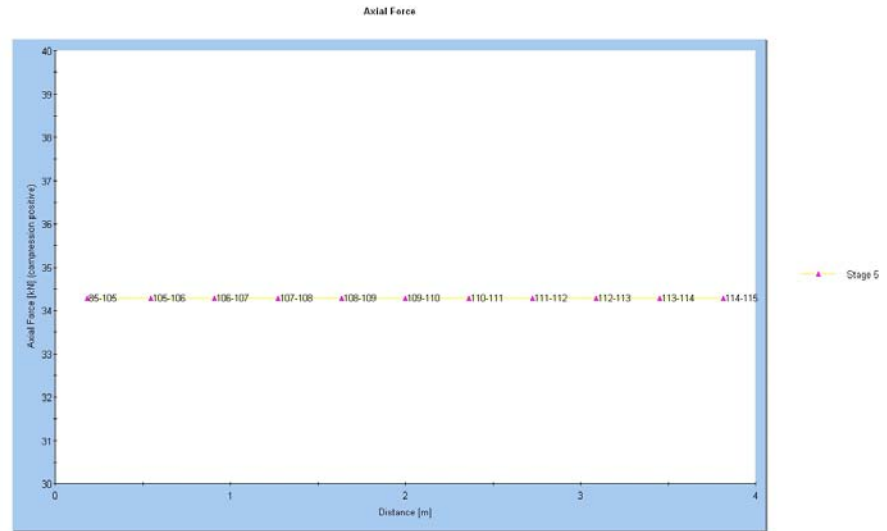


Now you can see even larger bending moments in the two soldier piles. Note the moment change from positive to negative in the right pile as it passes the strut, showing how the pile bends around the supporting strut. It is also interesting to observe how the left side of the trench is deformed to the left as it is being pushed by the strut.

The strut shows no moment because we set its moment of inertia to a very small value to prevent bending and twisting. To observe the effect of the strut, right click on it and select Graph Liner Data. Ensure that the Vertical Axis is set to Axial Force as shown.



Click Plot. You will now see a graph of the axial force in the strut that looks like this:



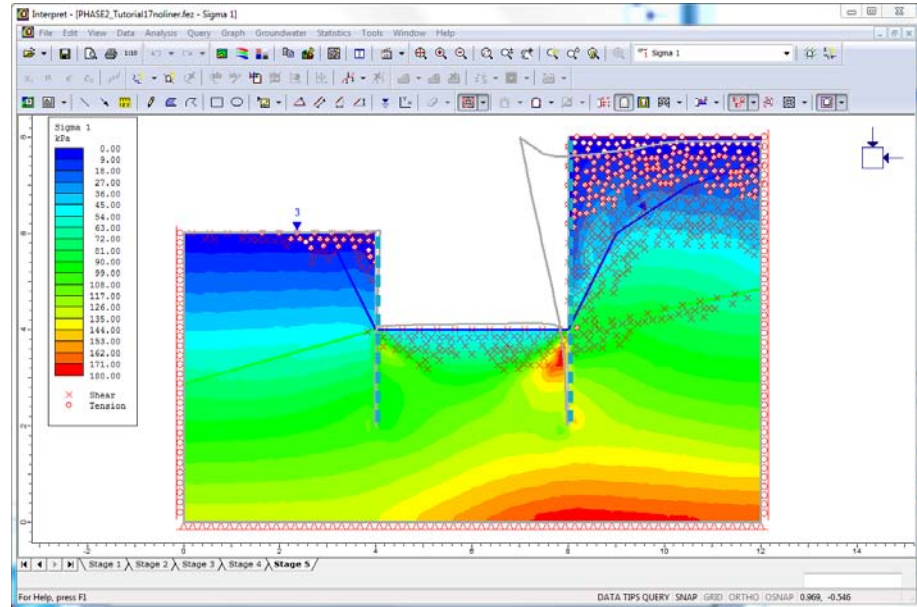
You can see the constant positive axial force experienced by the strut, clearly indicating that it is providing support to the excavation.

TIP: If you find the line hard to see, right click inside the graph and choose Chart Properties. Set the Chart Interior colour to grey. Alternatively, if you have Microsoft Excel software then you can right click on the graph and choose Plot in Excel.

This concludes the tutorial, you may now exit the *Phase2* Interpret and *Phase2* Model programs.

Additional Exercises

1. Try running the model again without the strut. You should see significantly more deformation and failure on the right side of the trench as shown below. Notice there is little deformation on the left side since the left side is not providing any support to the failing right side via the strut.



2. Try using structural interfaces for the soldier piles instead of liners. The structural interfaces should be composed of a Pile sandwiched between two joints. Set the joint Slip Criterion to Mohr-Coulomb and you should see slippage between the piles and the soil.